

Total No. of Questions : 8]

SEAT No. :

PD-4489

[Total No. of Pages : 2

[6403]-297

T.E. (Honors) (Cyber Security)
ENTERPRISE ARCHITECTURE AND COMPONENTS
(2019 Pattern) (Semester - VI) (310403)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right indicate full marks.
- 3) Draw neat diagram wherever necessary
- 4) Assume suitable data, if necessary.

- Q1)** a) Explain the significance of the component model in software architecture. [5]
b) Identify the key elements found in a component interaction diagram. [6]
c) How would you use a component description to document a software component? Provide an example. [6]

OR

- Q2)** a) Compare and contrast component interaction diagrams and component relationship diagrams. Provide examples for each. [9]
b) Design a component interaction diagram for a real-time messaging application, and explain your design choices. [8]

- Q3)** a) List three service qualities relevant to operational models. [6]
b) Explain the importance of understanding operational model design techniques in software development. [6]
c) Describe the role of service qualities in shaping operational model design decisions. [6]

OR

- Q4)** a) Explain the role of context in selecting operational model design techniques for a given project. Provide examples to support your explanation. [9]
b) Define the term 'operational pattern' and discuss its relationship with operational models. Provide examples to illustrate your explanation. [9]

P.T.O.

- Q5) a)** Provide definitions for ‘business scenario’ and ‘component deep dive’ in the context of software engineering. Explain how these terms are relevant to system design and development. [6]
- b)** Identify and explain three service qualities relevant to metadata management systems. Discuss their importance in ensuring the effectiveness of such systems. [6]
- c)** Define “metadata” and “master data” in the context of data management. Explain the importance of clear terminology in data-related projects. [5]

OR

- Q6) a)** Identify and explain three service qualities relevant to metadata management systems. Discuss their importance in ensuring the effectiveness and reliability of such systems. [8]
- b)** Describe a business scenario where effective metadata management is critical. Explain how proper metadata management practices can benefit the scenario. [9]
- Q7) a)** Describe how SABSA, COBIT, and TOGAF frameworks can be used to develop an enterprise security architecture. Provide a step-by-step approach, highlighting the key considerations at each stage. [9]
- b)** Explain how CMMI can be used to monitor, measure, and report the progress of architecture development activities within an organization. Discuss the key process areas and metrics relevant to this task. [9]

OR

- Q8) a)** Provide a real-life example of an organization that has successfully implemented SABSA, COBIT, and TOGAF frameworks in its enterprise architecture. Discuss the benefits achieved and challenges encountered during the implementation process. [9]
- b)** Outline the steps involved in using SABSA, COBIT, and TOGAF to develop an enterprise security architecture. Discuss the specific contributions of each framework in the process. [9]

